

Assignment 05

Class Diagram for University Timetable Optimization and Management System

Project: UTOS

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Assignment	Develop a detailed class diagram: classes, attributes, methods, multiplicities

Class Catalogue

The table below lists every class with its layer, key attributes, and key methods.

Class	Layer	Key Attributes	Key Methods
Person	Role	name: String, email: String	login(), logout(), getProfile()
Teacher	Role	employeeNumber: String, maxDailyLoad: Integer	submitAvailability(), requestChange(), viewPersonalTimetable()
Student	Role	registrationNumber: String	viewTimetable(), viewExamSchedule()
TimetableAdministrator	Role	authorityLevel: Integer	generateTimetable(), publishTimetable(), approveChangeRequest(), exportTimetable()
DepartmentCoordinator	Role	responsibilityArea: String	reviewTimetable(), approveChangeReq(), validateDeptData()
FacilityManager	Role	responsibilityArea: String	manageRoom(), viewUtilization(), updateRoomStatus()
SystemAdministrator	Role	(inherits Person)	manageUsers(), assignRole(), viewAuditLogs(), configureSystem()
University	Academic	name: String, campusName: String	getFaculties(): List, getDepartments(): List
Faculty	Academic	name: String	getDepartments(): List, getCourses(): List

Department	Academic	name: String, code: String	getTeachers(), getSections(), getCourses(), getTimetables()
AcademicTerm	Academic	name: String, startDate: Date, endDate: Date	getTimeslots(), getHolidays(), isActive(): Boolean
Course	Academic	code: String, title: String, creditHours: Integer	getOfferings(): List, getDepartment(): Department
StudentSection	Academic	name, program, semester, batch, size: Integer	getStudents(): List, getOfferings(): List
CourseOffering	Scheduling	weeklySessions: Integer, requiredRoomType: String, sessionDuration: Integer	getSessions(), getTeacher(), getSection(), getTerm()
Timetable	Scheduling	name, status: TimetableStatus, qualityScore: Float, hardConflictCount: Integer, unplacedSessionCount: Integer, createdAt: DateTime	generate(), publish(), getConflicts(), getQualityReport(), compareTo(), export(format)
ClassSession	Scheduling	sessionNumber: Integer, status: SessionStatus, locked: Boolean	assignRoom(room), assignTimeslot(ts), lock(), unlock(), getConflicts()
Timeslot	Scheduling	day: DayOfWeek, startTime: Time, endTime: Time, isMorning: Boolean, isLastSlot: Boolean	getDuration(): Integer, clashesWith(other): Boolean
TeacherAvailability	Scheduling	isAvailable: Boolean, note: String	getTimeslot(): Timeslot, getTeacher(): Teacher
SchedulingPolicy	Scheduling	name, policyType, priority: Integer, weight: Float, isHardConstraint: Boolean, isEnabled: Boolean	apply(session): Boolean, getPenalty(session): Float, validate(): Boolean
ChangeRequest	Scheduling	requestType, reason, urgency, status: RequestStatus, createdAt: DateTime	approve(note), reject(note), getSubmitter(): Person
Holiday	Scheduling	name: String, date: Date, reason: String	isOnDate(d: Date): Boolean
Building	Facility	name: String, location: String	getRooms(): List, getRoomCount(): Integer
Room	Facility	code, floor: Integer, capacity: Integer, roomType: RoomType, maintenanceStatus: Boolean	isAvailable(timeslot): Boolean, getUtilization(): Float
TimetableVersion	System	versionNumber: Integer, createdAt: DateTime, changeSummary: String	compareTo(other): VersionDiff, getChangedEntries(): List

Notification	System	message: String, notificationType, createdAt: DateTime, read: Boolean	markRead(), getRecipient(): Person
AuditLog	System	entityType, action, oldValue, newValue: String, timestamp: DateTime, performedBy: String	getActor(): Person

5. Relationships and Multiplicity

The following table documents every relationship in the class diagram with its type and multiplicity.

Relationship	Type	Multiplicity	Notes
Person → Teacher, Student, TimetableAdministrator, DeptCoordinator, FacilityManager, SystemAdministrator	Inheritance	1:1 (IS-A)	Person is abstract base; login/logout common to all
University → Faculty	Association	1 to 1..*	University has at least one Faculty
Faculty → Department	Association	1 to 1..*	Faculty has at least one Department
Department → Teacher	Association	1 to 0..*	Department employs zero or more Teachers
Department → StudentSection	Association	1 to 0..*	Sections belong to one Department
Department → Course	Association	1 to 0..*	Department owns courses it offers
Department → Timetable	Association	1 to 0..*	Department has its own timetables per term
AcademicTerm → CourseOffering	Association	1 to 0..*	Offerings are scoped to a term
AcademicTerm → Timeslot	Association	1 to 1..*	Term defines its available timeslots
AcademicTerm → Holiday	Association	1 to 0..*	Term may have zero or more holidays
AcademicTerm → Timetable	Association	1 to 0..*	Multiple timetable drafts per term allowed
Course → CourseOffering	Association	1 to 0..*	Course description reused across offerings
Teacher → CourseOffering	Association	1 to 0..*	Teacher may be assigned to multiple offerings
StudentSection → CourseOffering	Association	1 to 0..*	Section attends multiple course offerings

StudentSection → Student	Association	1 to 0..*	Section contains enrolled students
Teacher → TeacherAvailability	Association	1 to 0..*	Teacher may declare availability for many timeslots
TeacherAvailability → Timeslot	Association	1 to 1	Each availability record covers exactly one timeslot
Timetable → ClassSession	Composition	1 to 0..*	Sessions exist only within a timetable
CourseOffering → ClassSession	Association	1 to 1..*	At least one session per offering per week
ClassSession → Timeslot	Association	1 to 0..1	0..1 allows unplaced sessions without a timeslot
ClassSession → Room	Association	1 to 0..1	0..1 allows unplaced sessions without a room
Timetable → SchedulingPolicy	Association	1 to 0..*	Timetable is guided by active policies
Timetable → TimetableVersion	Association	1 to 1..*	Every saved timetable has at least one version
Building → Room	Composition	1 to 1..*	Rooms exist inside a building
FacilityManager → Room	Association	1 to 0..*	Manager is responsible for rooms
Person → ChangeRequest	Association	1 to 0..*	Any person can submit change requests
DeptCoordinator → ChangeRequest	Dependency	1 to 0..*	Coordinator reviews requests (behavioral coupling)
Timetable → Notification	Dependency	1 to 0..*	Publish/change events trigger notifications
TimetableAdministrator → AuditLog	Dependency	1 to 0..*	Admin actions are recorded in audit log

6. Enumerations

The following enumerations are used as attribute types throughout the class diagram.

Enumeration	Values
TimetableStatus	DRAFT, UNDER_REVIEW, PUBLISHED, ARCHIVED, REGENERATING
SessionStatus	PLACED, UNPLACED, LOCKED
RoomType	LECTURE_HALL, SEMINAR_ROOM, LAB, EXAM_ROOM, AMPHITHEATER
RequestStatus	PENDING, UNDER_REVIEW, APPROVED, REJECTED, WITHDRAWN
UrgencyLevel	LOW, MEDIUM, HIGH, CRITICAL
PolicyType	HARD_CONSTRAINT, SOFT_PREFERENCE

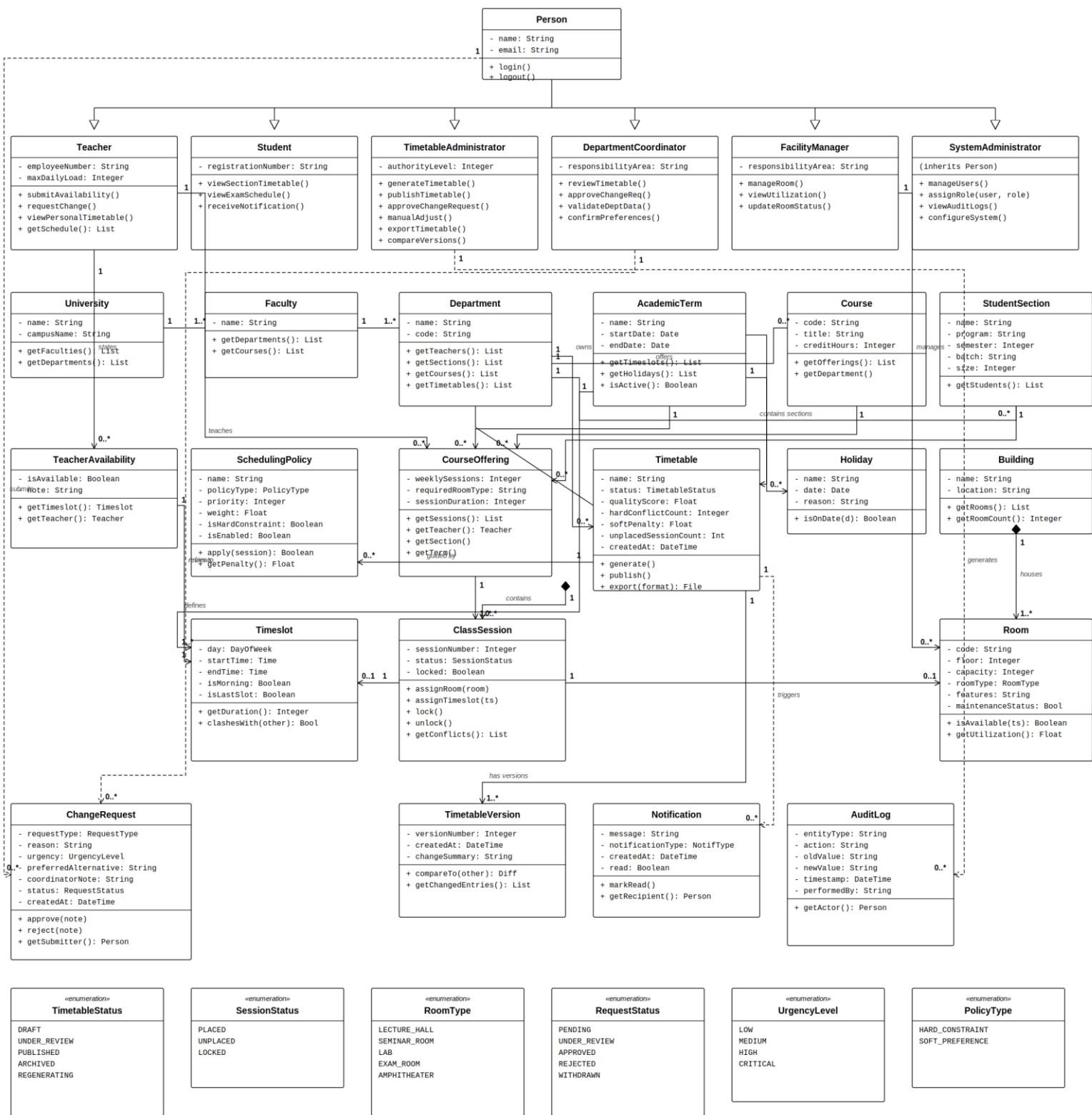
7. Key Design Decisions

Decision	Rationale
Person as abstract base	Centralises identity (name, email, login/logout) and avoids duplication across six actor types. All role-specific classes simply extend Person.
ClassSession 0..1 to Room and Timeslot	Allows unplaced sessions to exist as valid objects inside a draft timetable, supporting partial generation and conflict reporting.
CourseOffering separate from Course	Course is a reusable description; CourseOffering ties it to a specific term, teacher, and section. Same course can be offered multiple times across terms without duplication.
Timetable composes ClassSession	ClassSession only makes sense inside a Timetable. Composition signals that sessions are deleted if the timetable is removed.
TimetableVersion as separate class	Supports UC15 (Compare Timetable Versions) and provides an audit trail for timetable changes over time.
AuditLog as standalone class	Decouples logging from domain classes. Any entity change can be logged without modifying the entity itself.
SchedulingPolicy carries weight and isHardConstraint	Unifies hard and soft constraint configuration into one class, matching the constraint screen in the UI. isHardConstraint determines enforcement vs. penalisation.
Enumerations over raw Strings	TimetableStatus, SessionStatus, RoomType etc. prevent invalid attribute values and make the design implementation-ready.

8. Class Diagram

The full interactive class diagram is included in the interactive HTML submission and in the figure below. It is organised in six horizontal bands:

1. Roles and people — Person hierarchy at the top
2. Academic structure — University, Faculty, Department, AcademicTerm, Course, StudentSection
3. Scheduling core — CourseOffering, Timetable, ClassSession, Timeslot, TeacherAvailability
4. Policy and constraints — SchedulingPolicy, Holiday
5. Facility — Building, Room
6. System-wide — ChangeRequest, Notification, TimetableVersion, AuditLog, enumerations



UTOS Class Diagram — orthogonal routing, all multiplicities labeled